Parcial de Señales y Sistemas (66.74 y 86.05) (1era oportunidad)

27 de Octubre de 2014

Aclaración: Todos los desarrollos deben estar debidamente justificados. Aquellos desarrollos que no tengan una justificación clara en cada uno de sus pasos no serán válidos. Se pide también prolijidad y letra clara. Para aprobar es necesario probar conocimiento en todos los ejercicios. Si un ejercicio no se aborda el parcial no será aprobado aunque los otros ejercicios estén resueltos adecuadamente. Aclare en esta misma hoja nombre, padrón y práctica en la que cursa la materia.

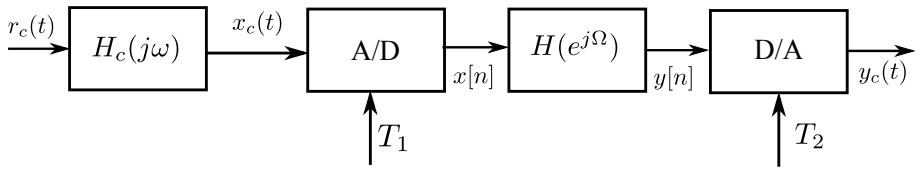
1. Considere un sistema LTI cuya respuesta al escalón vale:

$$\mathcal{T}\left[u[n]\right] = \frac{1}{n+1}u[n]$$

   a) Determine la respuesta al impulso del sistema.

   b) Analice la causalidad y estabilidad del mismo.

2. Considere el sistema de la figura donde $r_c(t) = \cos^2(\omega_0 t)\cos^2(2\omega_0 t)$ y los conversores A/D y D/A son ideales. El filtro $H_c(j\omega)$ es un filtro pasabajos ideal con frecuencia de corte $3\omega_0$. Considere además que $T_1 = \frac{2\pi}{3\omega_0}$.



   a) Asuma que $T_2 = T_1$ y que $H(e^{j\Omega}) = 1$ para todo $\Omega \in [-\pi, \pi]$. Determine la salida $y_c(t)$ del sistema y dibuje los espectros de cada una de las señales indicadas en la figura.

   b) Suponga que se desea que la salida del sistema sea $y_c(t) = \cos(2\omega_0 t)$. Diseñe un filtro $H(e^{j\Omega})$ adecuado y encuentre un valor de T_2 apropiado.

3. Considere un sistema LTI de tiempo discreto cuya respuesta al impulso vale

$$h[n] = \frac{\sin\left(\frac{3\pi n}{10}\right)}{\sin\left(\frac{\pi n}{10}\right)} \quad n = 0,\ \ldots\ , 9$$

   y cero en otro caso

   a) Suponga que la entrada al sistema vale:

$$x_1[n] = \frac{\sin\left(\frac{7\pi n}{10}\right)}{\sin\left(\frac{\pi n}{10}\right)} \quad n = 0,\ \ldots\ , 9$$

   y cero en otro caso. Considere la señal $\tilde{y}[n]$ que corresponde a la periodización (con periodo 10) de la salida $y[n]$ a la entrada $x_1[n]$. Determine $\tilde{y}[n]$.

   b) Determine la salida del sistema descripto por $h[n]$ cuando la entrada vale

$$x_2[n] = \cos\left(\frac{6\pi n}{10}\right) \text{ para todo } n$$

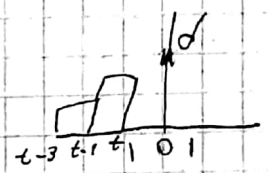
1ro 2° Cuat 2014

Dado un sist LTI

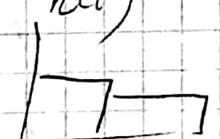
$$y(t)=\int_{T-1}^{t} x(\tau)\,d\tau+\int_{T-3}^{T-1}\frac{x(\tau)}{2}\,d\tau$$

a) $h(t)$ y justif.

$$h(t)=\int_{T-1}^{t}\delta(\tau)\,d\tau+\int_{T-3}^{T-1}\frac{\delta(\tau)}{2}\,d\tau$$



b) $h(t)=\begin{cases}0 & \text{si } t<0\\ 1 & \text{si } 0\le t\le 1\\ 1/2 & \text{si } 1<t<3\\ 0 & t>3\end{cases}$



además $\int_{-\infty}^{+\infty}|h(t)|dt<\infty$

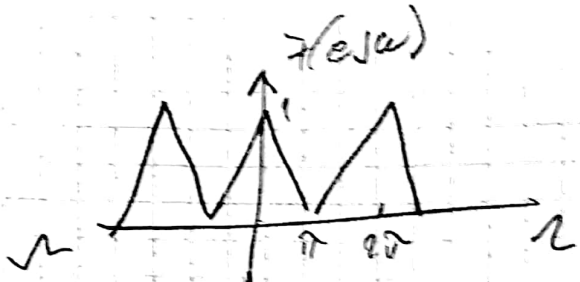
b) x par creciente en def

n

2) $x(n) \xrightarrow{x(n)} \boxed{D/A} \xrightarrow{x(t)} \boxed{h_c(t)} \xrightarrow{y_c(t)} \boxed{A/D} \xrightarrow{y(n)}$

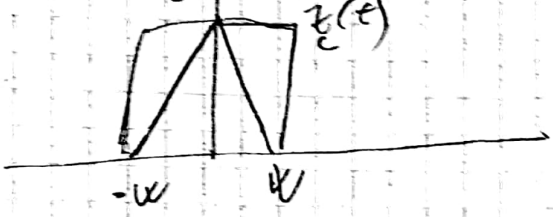
T_1

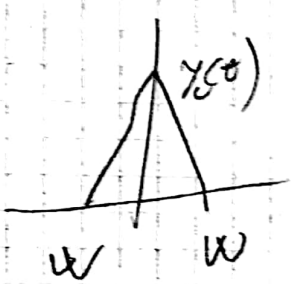
$$H_c(j\omega)=\begin{cases}1 & |\omega|<W\\ 0 & |\omega|>W\end{cases}$$

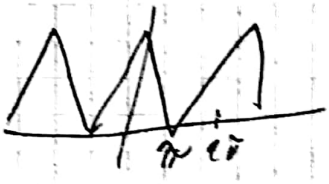


a) $y(n)$

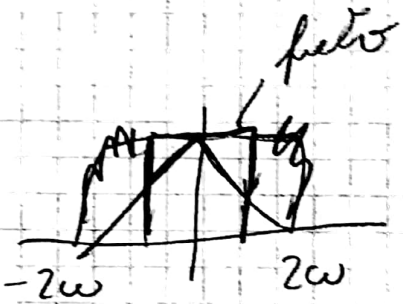
$T_1 = T_2$ π/ω



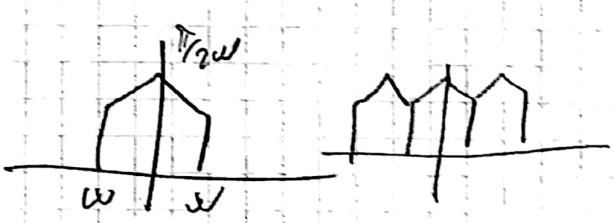




b) $T_1 = \frac{\pi}{2\omega}$



$T_2 = \frac{\pi}{\omega}$



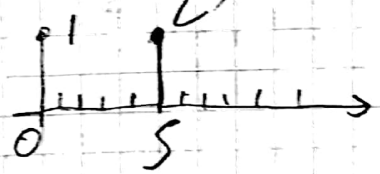


non lti

lti

③ $x(n) = \delta(n) + 2\delta(n-5)$

④ DFT longitud de $x(n)$



$X(k) = 1 + 2e^{-jk\frac{2\pi}{N}5}$

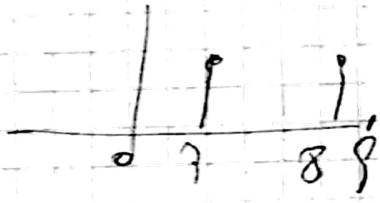
$= 1 + 2e^{-jk\pi}$

~~$X(k) = [1, 0, 0, 0, 0]$~~

Ecuación $y(n)$ / $y(k) = X(k) \cdot e^{j\frac{4\pi}{10}T}$

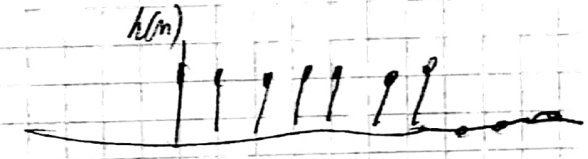
$e^{1 \cdot 2\pi/10 \cdot 2}$

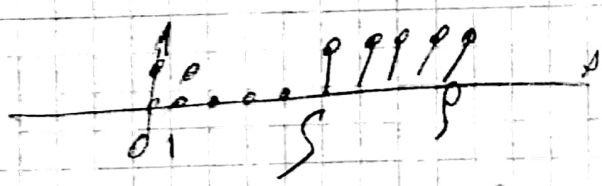
→ con 2 $\pi(k)$ [illegible]



c) $y(k) = X(k)H(k)$ / $H(k)$ = la DFT de

$h(n) = \begin{cases} 0 & 0 \le n \le 5 \\ 1 & 0 \text{ [illegible]} \end{cases}$







$$h_1(n) = A = \sum_{k=-\infty}^{n-1} \alpha^{n-k}\,\delta(k-\beta_1) \qquad \delta(k-\beta_1) = \begin{cases} 1 & k=\beta_1 \\ 0 & k\neq\beta_1 \end{cases}$$

Poiché in questa sommatoria solo il termine $\alpha^{n-\beta_1}$ se altro quando $k \geq \beta_1$

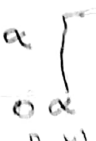
$\Rightarrow$ $n-1-\beta_1 \geq 0 \longrightarrow n-1 \geq \beta_1$

$$h_1(n) = \begin{cases} \alpha^{n-\beta_1} & n \geq \beta_1+1 \\ 0 & \text{altrimenti} \end{cases}$$

$$\hookrightarrow u(n) = \begin{cases} 1 & \forall n \geq 0 \\ 0 & \forall n < 0 \end{cases}$$



$$u(n-\alpha) = \begin{cases} 1 & \forall n-\alpha \geq 0 = n \geq \alpha \\ 0 & \forall n-\alpha \leq 0 = \end{cases}$$



$$U[n-\alpha] = \begin{cases} 1 & \text{se } n \geq \alpha \\ 0 & \text{c.v.c.} \end{cases}$$

$$h_1[n] = \alpha^{n-\beta_1} \cdot u[n-(\beta_1+1)]$$

$$h_2(n) = \gamma \sum_{k=-\infty}^{n-1} \alpha^{n-k}\,\delta(k-\beta_2) =$$

$n-1 \geq \beta_2$ $\quad \delta(k-\beta_2) \begin{cases} 1 & k=\beta_2 \\ 0 & k\neq\beta_2 \end{cases}$

Se all'interno del termine $\hookleftarrow$

$$h_2(n) = \begin{cases} \gamma\alpha^{n-\beta_2} & n \geq \beta_2+1 \\ 0 & n \leq \beta_2+1 \end{cases} \longrightarrow h_2(n) = \gamma\alpha^{n-\beta_2} \cdot u[n-(\beta_2+1)]$$

h_1

$$\alpha^n \cdot u(n) = \frac{1}{1-\alpha z^{-1}} \qquad |z| > |\alpha|$$

$$\alpha^{n-(\beta+1)} \cdot u(n-(\beta+1)) \overset{z}{=} \frac{z^{-(\beta+1)}}{1-\alpha z^{-1}}$$

$$\alpha^{n-\beta_1} \cdot \alpha \cdot u(n-(\beta+1)) \overset{z}{=} \frac{z^{-\beta+1)}}{1-\alpha z^{-1}}$$

$$h_1(n) = \alpha^{n-\beta_1} \cdot u(n-(\beta_1+1)) \overset{z}{=} \frac{\alpha z^{-(\beta_1+1)}}{1-\alpha z^{-1}}$$

$$h_2(n) = \gamma\, \alpha^{n-\beta_2} \cdot u(n-(\beta_2+1)) \overset{z}{=} \frac{\gamma \cdot \alpha z^{-(\beta_2+1)}}{1-\alpha z^{-1}}$$

$$H(z) = \frac{\alpha z^{-\beta_1-1}}{1-\alpha z^{-1}} + \frac{\gamma \alpha z^{-\beta_2-1}}{1-\alpha z^{-1}}$$

$$= \frac{\alpha z^{-1}\left(z^{-\beta_1} + \gamma z^{-\beta_2}\right)}{1-\alpha z^{-1}}$$

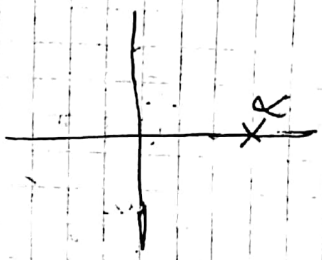
$$z^{-\beta_1} + \gamma z^{-\beta_2} = 0$$

$$z^{-\beta_1} = -\gamma z^{-\beta_2}$$

$$\frac{z^{-\beta_1}}{z^{-\beta_2}} = -\gamma$$

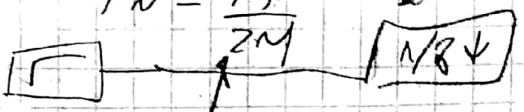
$$z^{-\beta_1+\beta_2} = -\gamma$$

$$\sqrt[-\beta_1+\beta_2]{-\gamma} = z$$



mscaremel@gmail.com.

Diezmar $f_N = \frac{f_s}{2M}$ (de)



(Wn 8 para 44.1)

$f_s/8$

$U(jw) = \int_0^1$

$\frac{w_c}{M} < \pi$

88,2 = m orden

$W_N = \Omega_N \cdot f_s$

$T_s = 1/f_s$

$w_N = \frac{\Omega_N}{f_s} < \pi/M$

Plot(x,y)

grid on

Eff. 2N

Volpeps $= \frac{M}{2} \cdot 1/f_s$

$2w_M \leq w_s$

$w_M \leq w_s/2 = 22k$

$w_s = 2\pi \cdot f_s$

con Jandelio

w_{Mex}

$f_s = 44100$

$T_s = 1/44100$

$T_s = 1/44100$

0 1 wpi

w_s w_s

M=8

$-\pi$ π

1

$8w_M$ $8w_M$

$\frac{f_s}{2N}$

$2w_{Mex} > 8w_s$

22k si diezmo

$8w_{Mex} \leq 22$

$\pi/8$

$f_s/8$

2

HOJA Nº

27/10/14

1) LTI

$$T[u[n]] = \frac{1}{n+1}\, u[n]$$

a) $h[n]$

b) analizar la causalidad y estabilidad del mismo

$$h[n] = T[u[n] - u[n-1]]$$

$$T[u[n] - u[n-1]] = \frac{1}{n+1} u[n] - \frac{1}{n-1+1} u[n-1]$$

$$h[n] = \frac{1}{n+1} u[n] - \frac{1}{n} u[n-1]$$

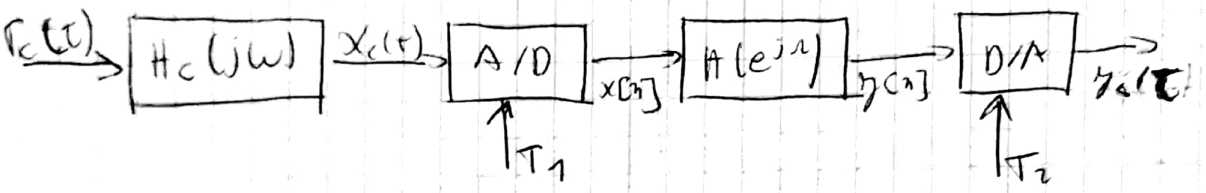
b) estabilidad $\sum_{n=-\infty}^{\infty} |h[n]| < \infty$

$$\sum_{0}^{\infty} \frac{1}{n+1} - \underbrace{\sum_{1}^{\infty} \frac{1}{n}}_{\text{Diverge}} \Rightarrow \text{no es estable}$$

- Causalidad → que empiece en 0 ⇒ Sí

27/10/14

2) Considere el sistema de la figura donde $r_c(t) = \cos^2(W_0 t)\cos^2(2W_0 t)$ y los conversores A/D y D/A son ideales. El filtro $H_c(jW)$ es un filtro pasabajos ideal con frecuencia de corte $3W_0$. Considere además que $T_1 = \frac{2\pi}{3W_0}$

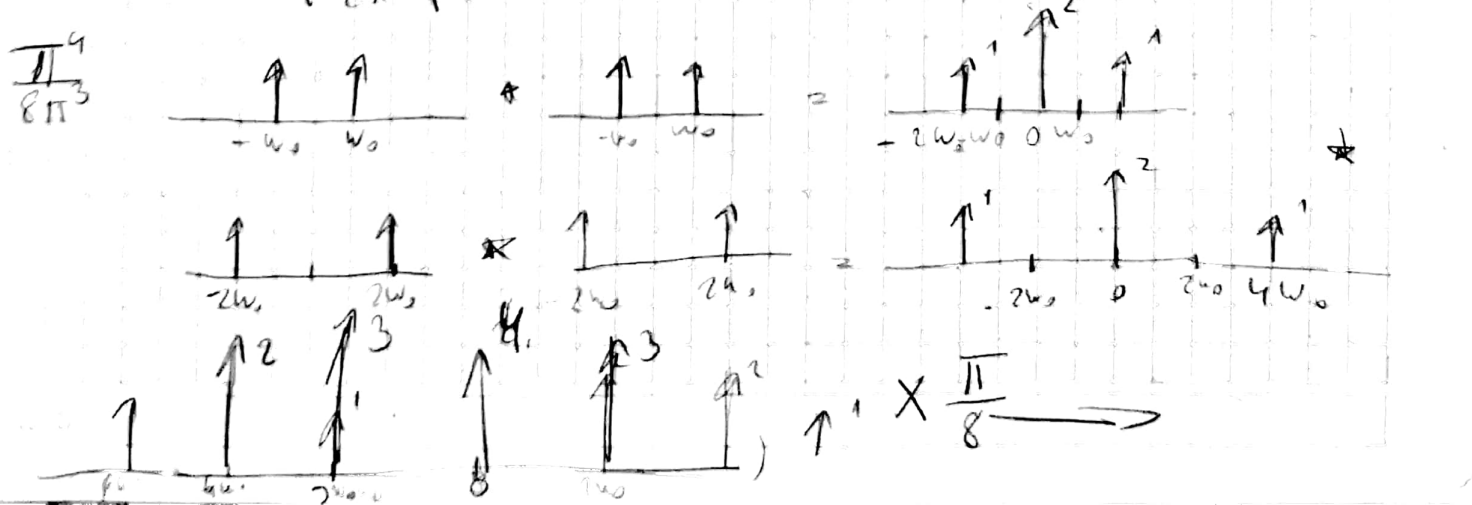


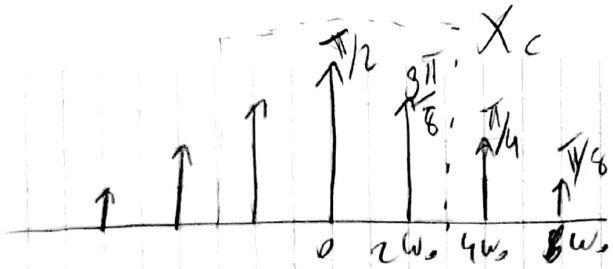
a) Asuma que $T_2 = T_1$ y que $H(e^{j\Omega}) = 1 \ \forall \Omega \in [-\pi, \pi]$. Determine la salida $y_c(t)$ del sistema y dibuje los espectros de cada una de las señales indicadas en la figura

b) Suponga que se desea que la salida del SIST sea $y_c(t) = \cos(2W_0 t)$. Diseñe un filtro $H(e^{j\Omega})$ adecuado y encuentre un valor de T_2 apropiado

a) $\Rightarrow \frac{1}{2\pi}\left(\frac{1}{2\pi}\left(\pi \sum \left(\delta(\Omega - \Omega_0) + \delta(\Omega + \Omega_0)\right) * \left(\pi \sum \left(\delta(\Omega - \Omega_0) + \delta(\Omega + \Omega_0)\right)\right) *\right.\right.$

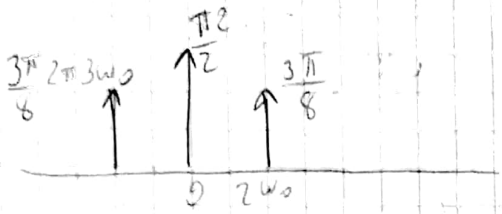
$* \left(\frac{1}{2\pi}\left(\pi \sum \left(\delta(\Omega - 2\Omega_0) + \delta(\Omega + 2\Omega_0)\right) * \left(\pi \sum \left(\delta(\Omega - 4\Omega_0) + \delta(\Omega + 4\Omega_0)\right)\right)\right.\right.$





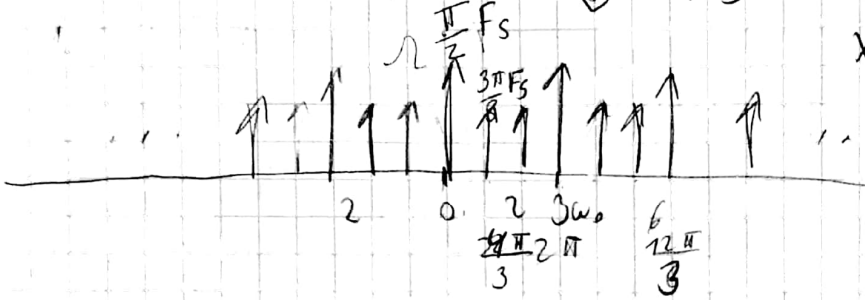
Filtro $\omega_s = 3\omega_0$

$X_c(\omega)$



$T_s = \frac{2\pi}{3\omega_0} \Rightarrow \omega_s = 3\omega_0$

↓ A/D



$\Omega = \omega T_1$
$= \omega \frac{2\pi}{3\omega_0}$

$Y(\Omega) = \underbrace{H(\Omega)}_{1} X(\Omega)$

$T_2 = \frac{2\pi}{3\omega_0}$

(5) 14/60

$X_p(j\omega) = X(e^{j\Omega})\Big|_{\Omega=\omega T} = X(e^{j\omega T})$

$X(\Omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c\left(j\left(\frac{\Omega}{T} - \frac{2\pi k}{T}\right)\right)$

$X(e^{j\Omega}) = \frac{1}{T} X_c\left(j\frac{\Omega}{T}\right)$

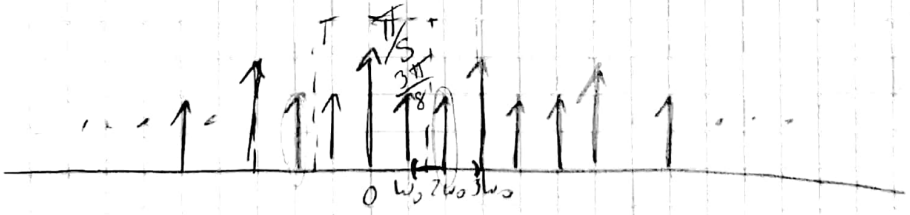
$H_c($

(5) 9/60 Pazljivo $\omega \in (-\pi, \pi)$

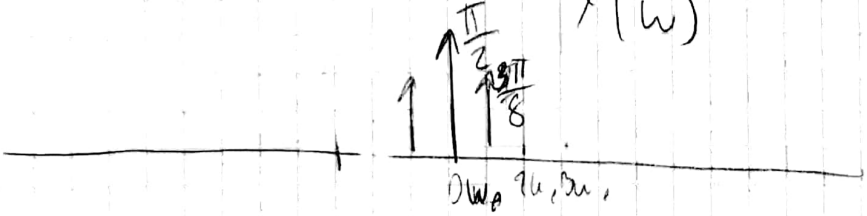
$\omega < \omega_c < \omega_s - \omega$

$\omega_0 < \omega_c < \frac{2\pi}{3\omega_0} - \omega_0$

$\omega_0 < \omega_c < 2\omega_0$



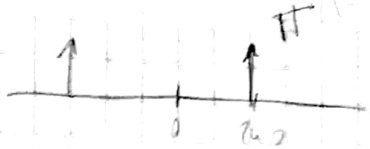
$X(\omega)$



$y_c(t) = \cos(\omega_0 t)\frac{3}{8} + \frac{1}{4}$

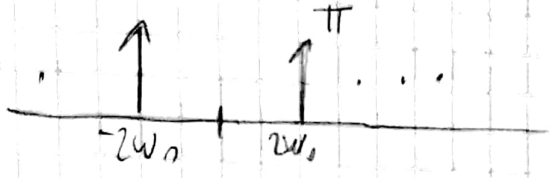
27/10/14

b) $y[n] = (\cos 2\omega_0 n)$



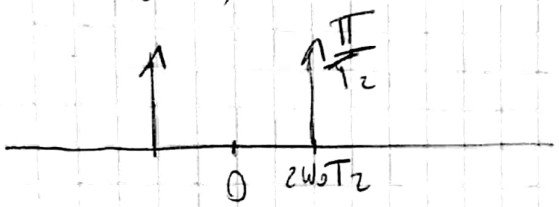
Busco $Y(j\omega)$

$Y_p(j\omega)$



$\Omega = \omega T_z$

$Y(\Omega)$



$Y[\Omega] = H[\Omega]\, X[\Omega]$

$\Rightarrow H[\Omega]$ es un filtro pasabanda de dos apuntes con frecuencia central ω_0 o $2\omega_0$, que el ancho de esta banda sea menor a ω_0.

También se quiere que la ganancia sea de $\frac{8}{3}\frac{T_z}{T_1}$ para que quede únicamente el π

27 de Octubre 2014

① Sistema LTI. $T[u[n]] = \frac{1}{n+1} u[n]$

a)

$u[n] * h[n] = \frac{1}{n+1} u[n]$ $\quad T[\delta(n)] = h[n]$

Por ser un sistema LTI, lineal e invariante en el tiempo, se cumple que

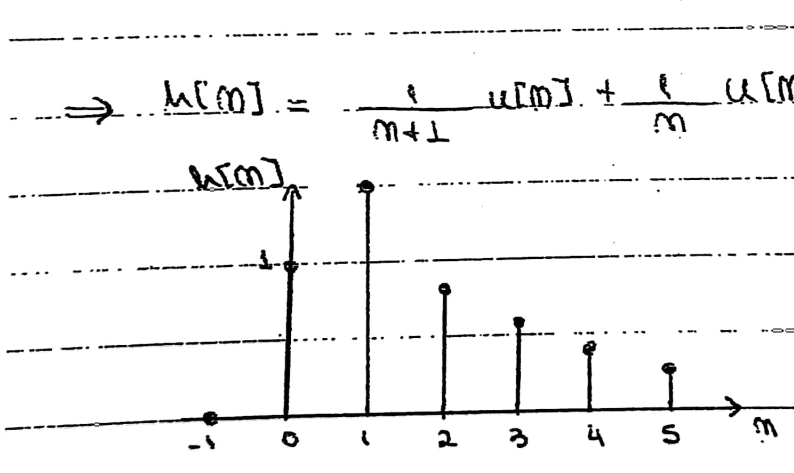
$$\left.\begin{array}{l} x \longrightarrow \boxed{H} \longrightarrow x' \\ y \longrightarrow \boxed{H} \longrightarrow y' \end{array}\right\} \Rightarrow \alpha x + y \longrightarrow \boxed{H} \longrightarrow \alpha x' + y'$$

Para obtener la respuesta al impulso ($\delta(n)$) se puede escribir como: $\delta(n) = u(n) - u(n-1)$

$(u[n] - u[n-1]) * h[n] = u[n] * h[n] + u[n-1] * h[n] =$

$= \frac{1}{n+1} u[n] + \frac{1}{n} u[n-1]$

$$\Rightarrow h[n] = \frac{1}{n+1} u[n] + \frac{1}{n} u[n-1] \begin{cases} 0 & n < 0 \\ \frac{1}{n+1} & 0 \le n < 1 \\ \frac{1}{n+1} + \frac{1}{n} & n \ge 1 \end{cases}$$



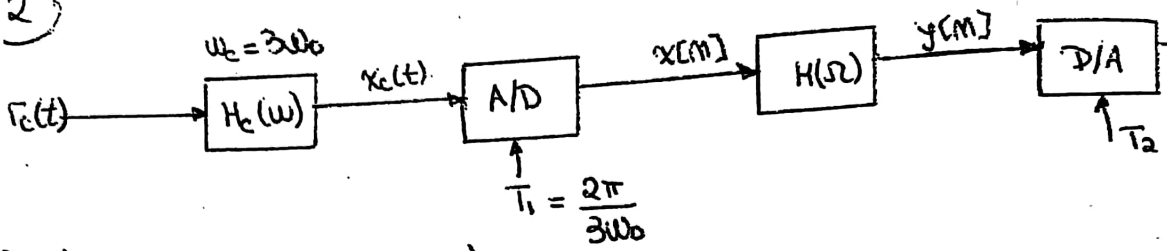
b) Se puede ver que el sistema no es estable ya que: $\sum_{-\infty}^{\infty} |h(n)| \to \infty$ porque $\sum_{\infty}^{\infty} \frac{1}{n}$ es divergente

Además, como $h(n) = 0 \quad n < 0 \Rightarrow y(n) = \sum_{k=-\infty}^{\infty} x[k]\, h[n-k] =$

$= \sum_{n=-\infty}^{0} x[k]\, h[n-k]$ es un sistema causal porque no necesita información del futuro para resolver su salida.

Imagen en papel Personal 80g/m² de Calculadora Arrastrada

2)



$r_c(t) = \cos^2(w_0 t)\cos^2(2w_0 t)$

1) $T_2 = T_1$, $H(\Omega) = 1 \quad \forall \Omega \in [-\pi, \pi]$

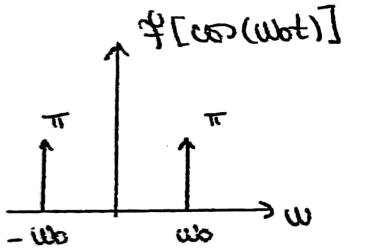
$r_c(t) = \cos(w_0 t)\cos(w_0 t)\cos(2w_0 t)\cos(2w_0 t)$

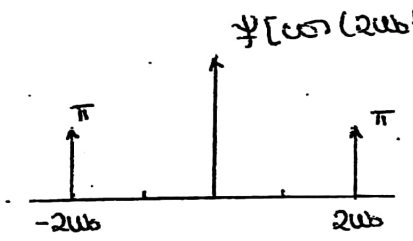
$R_c(w) = \mathcal{F}[\cos(w_0 t)\cdot\cos(w_0 t)\cdot\cos(2w_0 t)\cos(2w_0 t)] =$

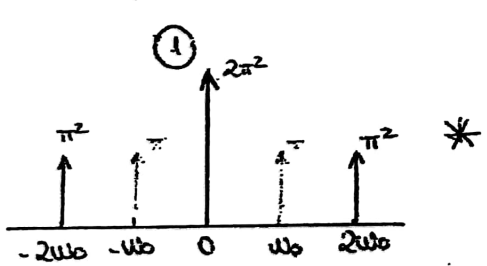
$= \frac{1}{2\pi}\mathcal{F}[\cos(w_0 t)\cos(w_0 t)] * \mathcal{F}[\cos(2w_0 t)\cos(2w_0 t)] =$

$= \frac{1}{2\pi}\cdot\frac{1}{2\pi}\mathcal{F}[\cos(w_0 t)] * \mathcal{F}[\cos(w_0 t)] * \frac{1}{2\pi}\mathcal{F}[\cos(2w_0 t)] * \mathcal{F}[\cos(2w_0 t)] =$

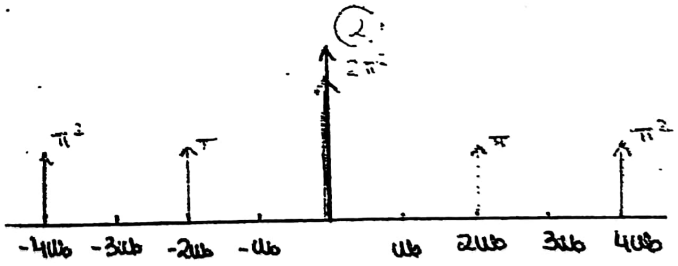
$= \frac{1}{8\pi^3}\underbrace{\mathcal{F}[\cos(w_0 t)] * \mathcal{F}[\cos(w_0 t)]}_{①} * \underbrace{\mathcal{F}[\cos(2w_0 t)] * \mathcal{F}[\cos(2w_0 t)]}_{②}$ ③

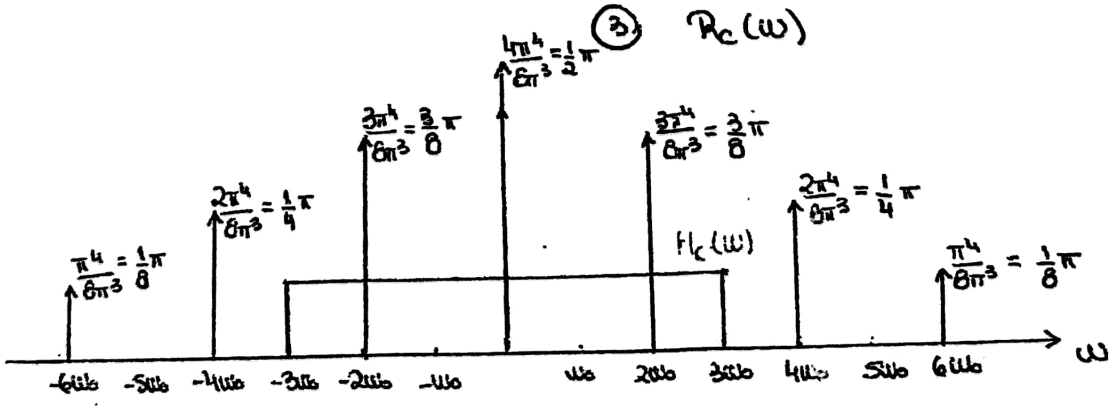


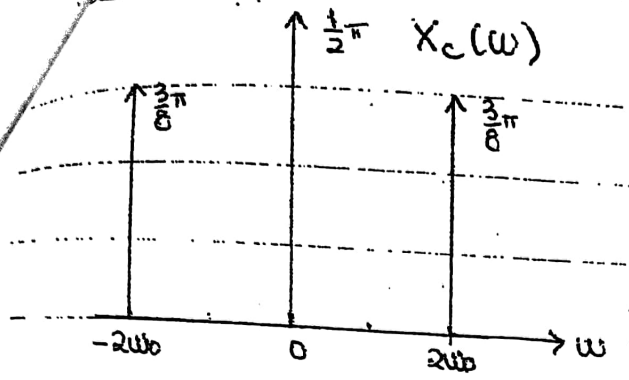




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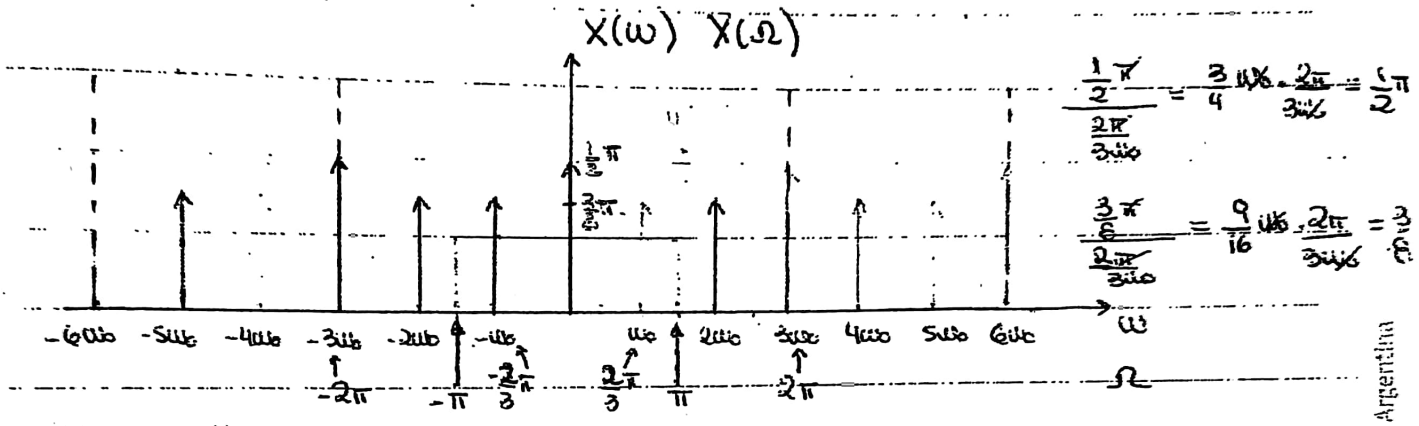






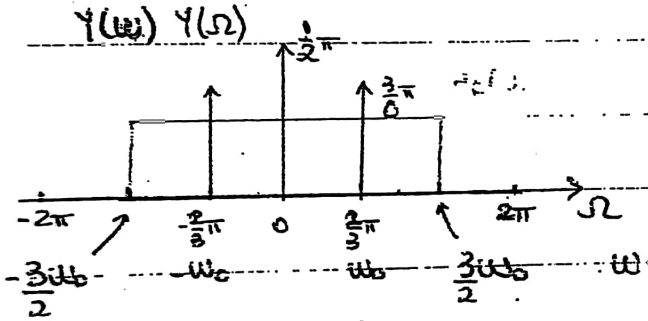
$T_1 = \frac{2\pi}{3\omega_0} \Rightarrow \omega_s = 3\omega_0$

$\Omega = \omega_s T_s = 3\omega_0 \cdot \frac{2\pi}{3\omega_0} = 2\pi$



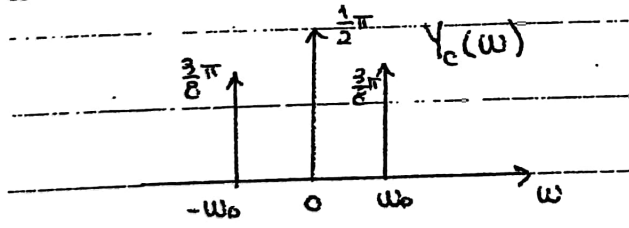
$\frac{\frac{1}{2}\pi}{\frac{2\pi}{3\omega_0}} = \frac{3}{4}\omega_0 \cdot \frac{2\pi}{3\omega_0} = \frac{1}{2}\pi$

$\frac{\frac{3}{8}\pi}{\frac{2\pi}{3\omega_0}} = \frac{9}{16}\omega_0 \cdot \frac{2\pi}{3\omega_0} = \frac{3}{8}\pi$



como $T_2 = T_1 \Rightarrow \frac{\Omega}{T} = \omega_s$

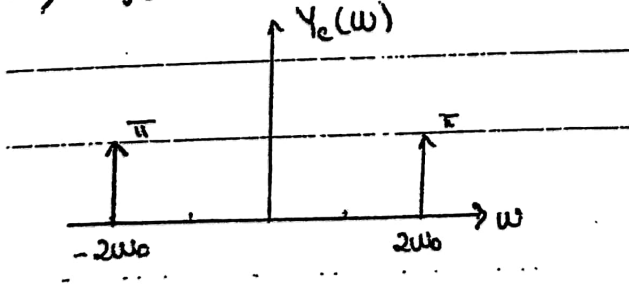
$\omega_s = \frac{2\pi}{\frac{2\pi}{3\omega_0}} = \omega_0$

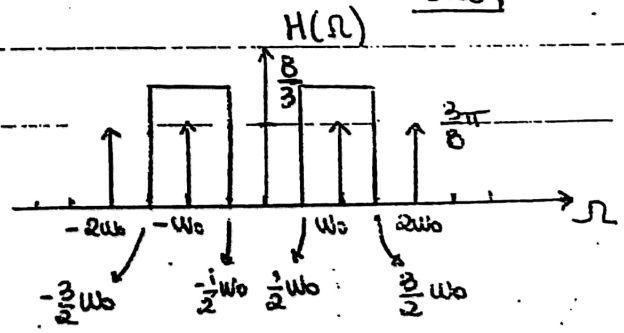


$y(t) = \frac{1}{2}\cos^2\left(\frac{\omega_0}{2}t\right) + \frac{1}{2}\cos(\omega_0 t)$

b) $y_c(t) = \cos(2\omega_0 t)$

$\frac{2}{3}\pi = \omega_s \cdot T_2 \Rightarrow \boxed{T_2 = \frac{\frac{2}{3}\pi}{2\omega_0} = \frac{\pi}{3\omega_0}}$ ($2\omega_0$)

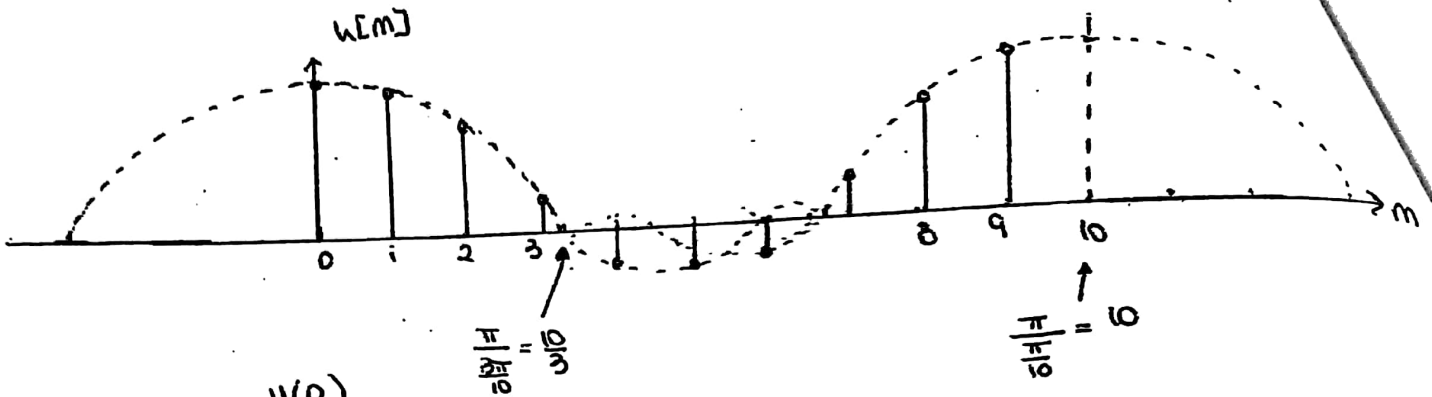


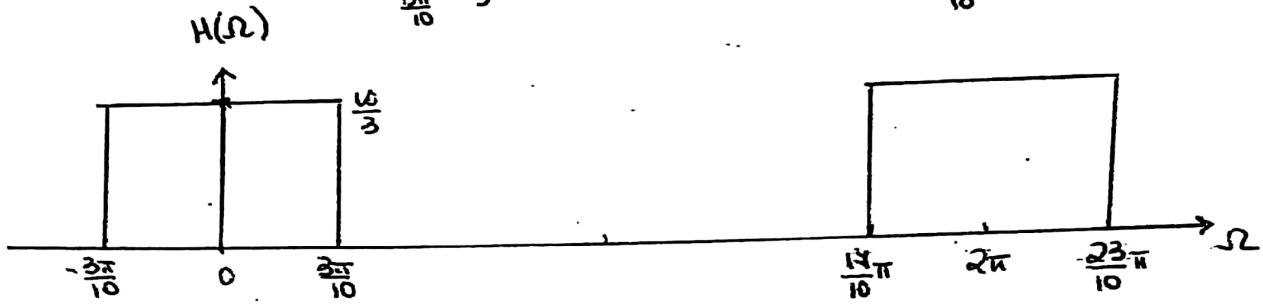


Notas:

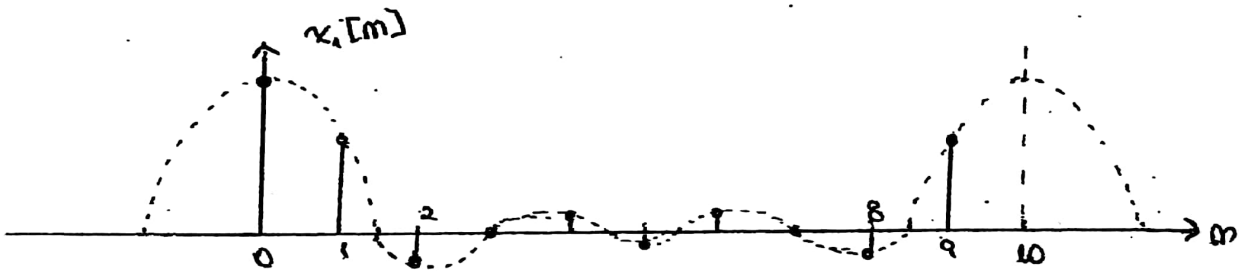
Impreso en papel Boreal Cel 80g/m2 de Celulosa Argentina

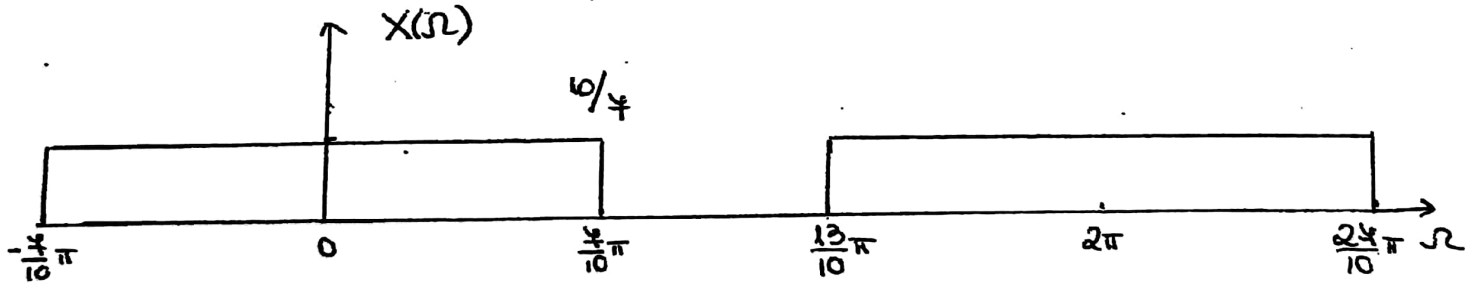
3) LTI $\quad h[m] = \dfrac{\text{sen}\left(\frac{3\pi}{10}m\right)}{\text{sen}\left(\frac{\pi}{10}m\right)} \quad m = 0, \ldots, 9$



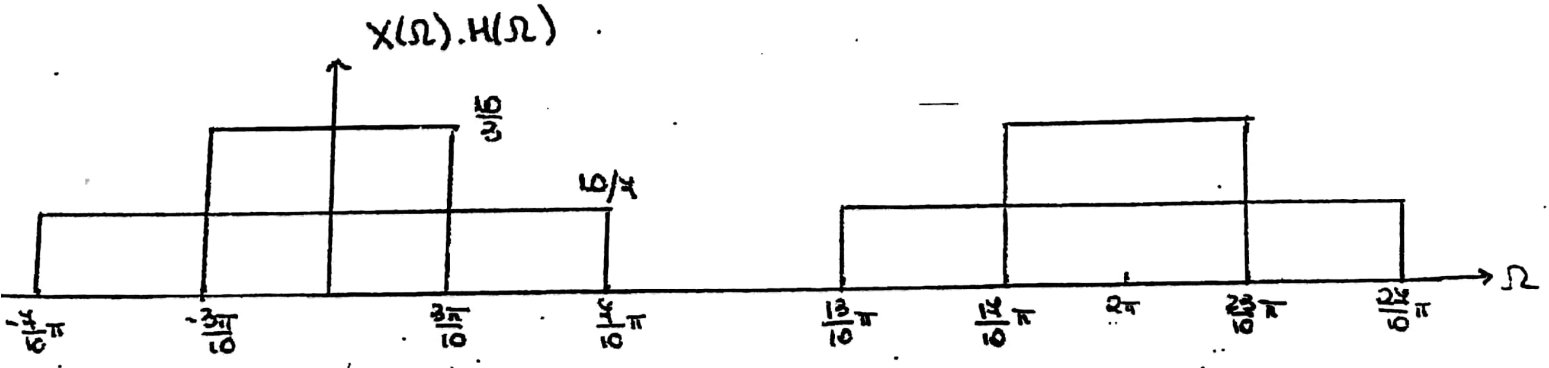


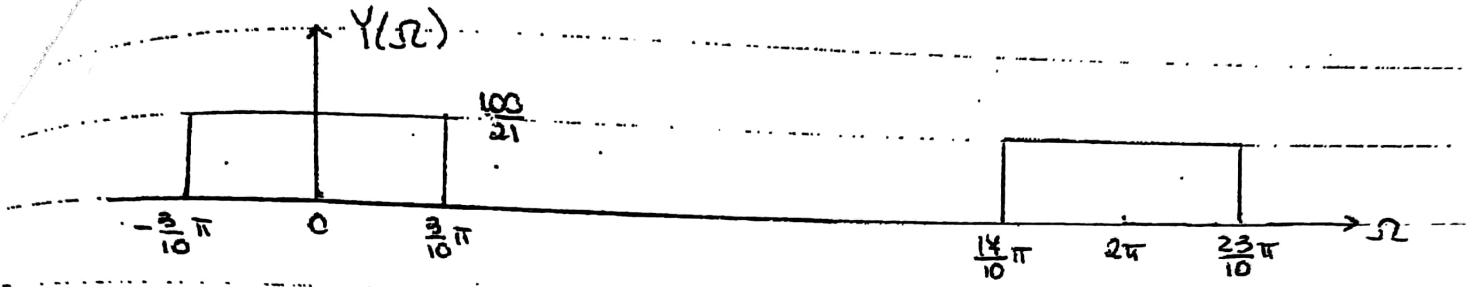
a) $x_1[m] = \dfrac{\text{sen}\left(\frac{4\pi}{10}m\right)}{\text{sen}\left(\frac{\pi}{10}m\right)} \quad m = 0, \ldots, 9$

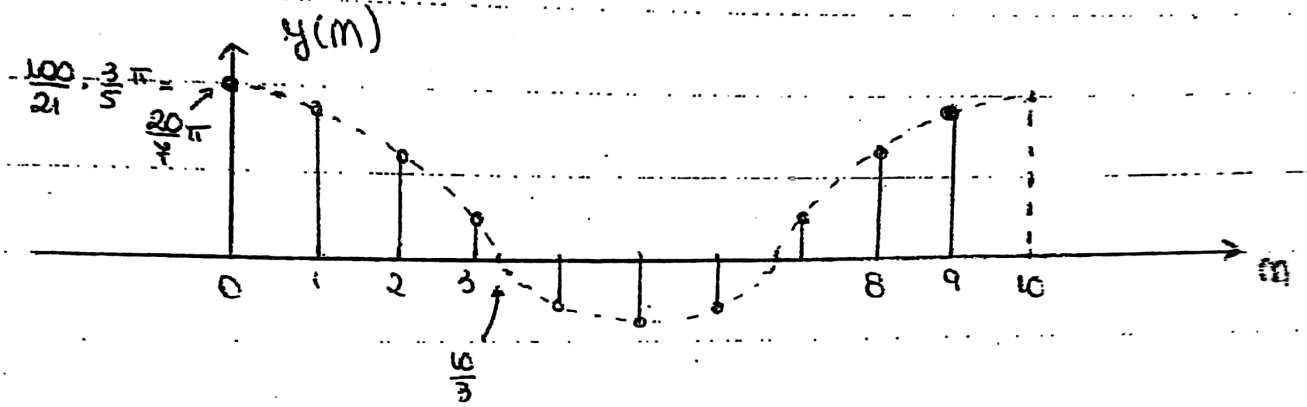




$\tilde{x}[m] \circledast \tilde{h}[m] \longrightarrow a_k[k]\,b_k[k]$

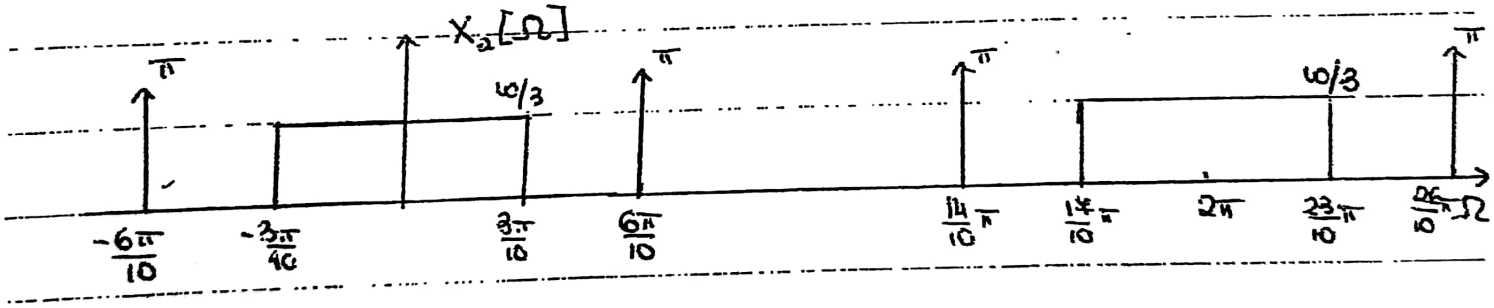






$$y[m] = \frac{20\pi}{4}\,\frac{\operatorname{sen}\left(\frac{3}{10}\pi m\right)}{\operatorname{sen}\left(\frac{\pi}{10} m\right)} \qquad m = 0, \ldots, 9$$

b) $x_2[m] = \cos\left(\frac{6\pi m}{10}\right) \quad \forall\, m$



$y[m] = 0 \qquad m = 0, \ldots, 9$

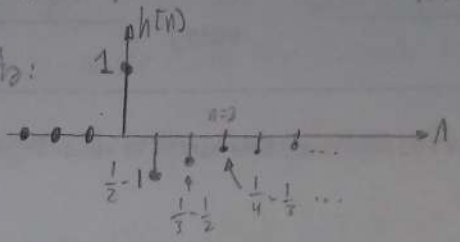
Impreso en papel Boreal cal 80g/m2 de Celulosa Argentina

(2/10/14)

1.2) Me dicen que la respuesta de un sist. LTI al escalón $u[n]$ es $s[n] = \frac{u[n]}{n+1}$ y me piden con eso hallar la respuesta al impulso $h[n]$.

Recordando que $h[n] = s[n] - s[n-1]$, puedo directamente plantear que $h[n] = \frac{u[n]}{n+1} - \frac{u[n-1]}{n}$

Es decir, $h[n]$ es una señal con la sgte pinta:



b) Recordando las sgtes dos propiedades:

I) El sist. es causal si $h[n] = 0 \ \forall n < 0$

II) El sist. es estable si: $\sum_{n=-\infty}^{+\infty} |h[n]| < \infty$

Se puede ver rápidamente que el sist. es no causal (pues $h[0] = 1$) y es estable (pues $h[n]$ es tal que $\sum_{k=-\infty}^{+\infty} |h[n]| = \sum_{k=-\infty}^{\infty} \left| \frac{u[n]}{n+1} - \frac{u[n]}{n} \right| = \sum_{k=1}^{+\infty} \left| \frac{1}{n+1} - \frac{1}{n} \right| + 1$ y $\sum_{k=1}^{+\infty} \left| \frac{1}{n+1} - \frac{1}{n} \right| \leq \sum_{k=1}^{\infty} \frac{1}{n+1} + \sum_{k=1}^{+\infty} \frac{1}{n} < \infty$.)

(1/01/14)

1.2) Me dicen que la respuesta $y(t)$ de un sist. lineal es $y(t) = e^{-\alpha t}$ cuando $x(t) = e^{\alpha t}$. Con eso, me piden hallar $y(t)$ cuando $x(t)$ sea par o impar. Para ello, considero la serie de Fourier de la señal $x(t)$, pues sé que $x(t)$ es periódica con período T. $\Rightarrow$ Puedo escribir a $x(t)$ como $x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{j\omega k t}$ donde $\omega = \frac{2\pi}{T}$ y a_k son los coef. de Fourier asociados.

$\Rightarrow$ Como el sist. es lineal, $y(t) = T[x(t)] = T\left[\sum_{k=-\infty}^{+\infty} a_k e^{j\omega k t}\right] = \sum_{k=-\infty}^{+\infty} a_k T[e^{j\omega k t}]$

$\Rightarrow y(t) = \sum_{k=-\infty}^{+\infty} a_k e^{-j\omega k t} \ \forall x(t)$ periódica. Para posterior conveniencia, reescribo a $y(t)$ como:

$y(t) = \sum_{k=1}^{+\infty} a_k e^{-j\omega k t} + \sum_{k=-\infty}^{-1} a_k e^{-j\omega k t} + a_0$. Veo ahora los dos casos ($x(t)$ par o impar):

I. Si $x(t)$ es par $\Rightarrow a_k = a_{-k} \Rightarrow y(t) = \sum_{k=1}^{+\infty} a_k e^{-j\omega k t} + \sum_{k=1}^{+\infty} a_k e^{j\omega k t} + a_0$

$\Rightarrow y(t) = \sum_{k=1}^{+\infty} a_k \left(e^{j\omega k t} + e^{-j\omega k t}\right) + a_0 = 2\sum_{k=1}^{+\infty} a_k \left(\frac{e^{j\omega k t} + e^{-j\omega k t}}{2}\right) + a_0 = a_0 + 2\sum_{k=1}^{+\infty} a_k \cos(\omega k t)$

$\Rightarrow$ Si $x(t)$ es par, $y(t) = a_0 + 2\sum_{k=1}^{+\infty} a_k \cos(\omega k t)$ donde a_k son los coef. de Fourier

II. Si $x(t)$ es impar $\Rightarrow a_k = -a_{-k} \Rightarrow y(t) = \sum_{k=1}^{+\infty} a_k e^{-j\omega k t} - \sum_{k=1}^{+\infty} a_k e^{j\omega k t} + a_0$

$\Rightarrow y(t) = \sum_{k=1}^{+\infty} a_k \left(e^{-j\omega k t} - e^{j\omega k t}\right) + a_0 = 2j \sum_{k=1}^{+\infty} -a_k \left(\frac{e^{j\omega k t} - e^{-j\omega k t}}{2j}\right) + a_0$

$\Rightarrow y(t) = a_0 - 2j \sum_{k=1}^{+\infty} a_k \operatorname{sen}(\omega k t)$ con $x(t)$ impar y a_k los coef. de Fourier.

b) Sea $x(t) = e^{+\alpha(t-t_0)}$. Luego, como $T[e^{\alpha t}] = e^{\alpha t} \Rightarrow$ para que el sist. sea ~~lineal~~ inv., la respuesta a $x(t)$ debe ser $y(t) = e^{-\alpha(t-t_0)}$ pues si $T[x(t)] = b(t) \Rightarrow T[x(t-t_0)] = b(t-t_0)$.

Ahora, como $x(t) = e^{\alpha(t-t_0)} = \frac{e^{\alpha t}}{e^{\alpha t_0}} \Rightarrow T[x(t)] = \frac{T[e^{\alpha t}]}{e^{\alpha t_0}} = \frac{e^{\alpha t}}{e^{\alpha t_0}} = e^{-\alpha(t+t_0)}$

$\Rightarrow$ El sist. no es lineal, pues quedó que $y(t) = e^{-\alpha(t+t_0)}$ en lugar de $e^{-\alpha(t-t_0)}$.